

Classifying Real vs. Fake News: A Machine Learning Approach

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Abstract

This study serves as an experimental baseline to evaluate whether traditional machine learning approaches can effectively classify legitimate versus fabricated news using a dataset of 150,000 articles. Initial exploratory data analysis revealed that, contrary to common expectations, fake news articles in this dataset rely on longer content bodies rather than purely sensationalist headlines. To optimize text classification, an advanced cleaning pipeline was developed that merged article titles with body content to preserve crucial vocabulary. Vectorization techniques (Bag of Words, TF-IDF, and Word2Vec) were evaluated alongside three baseline classification models (Logistic Regression, Linear SVM, and Random Forest). While TF-IDF provided the clearest visual class separation during Principal Component Analysis (PCA), Word2Vec achieved higher empirical accuracy. The strongest predictive outcome peaked at 55.53% accuracy using Word2Vec and Logistic Regression. Ultimately, this experiment demonstrates that while traditional machine learning models can capture basic textual patterns, their limited accuracy indicates that more advanced deep learning architectures are likely required for highly reliable fake news detection.

Introduction

The rapid spread of misinformation online has made the automatic detection of fake news a critical challenge in modern data science (Shu et al., 2017). As deceptive articles become increasingly sophisticated, relying on manual verification is no longer sufficient. While state-of-the-art solutions often utilize complex deep learning models (such as Neural Networks or Transformers), this study serves as an experimental baseline to test the viability and limits of traditional machine learning approaches for this complex natural language task.

By investigating a dataset of 150,000 articles, this project evaluates the impact of foundational data cleaning strategies and vectorization methods—specifically Bag of Words, TF-IDF, and Word2Vec. Furthermore, it benchmarks the performance of Logistic Regression, Linear SVM, and Random Forest algorithms. The primary goal of this experiment is to determine whether these traditional computational methods can effectively identify fabricated news, or if the nuanced complexity of human deception strictly requires the more robust, context-aware capabilities provided by deep learning networks (Chollet, 2026).

I. Dataset Overview and Exploratory Analysis

1.1 Dataset Characteristics

The dataset utilized for this study comprises news articles collected to distinguish between legitimate and fabricated news. Each entry represents a single article containing a title, the main body of content, and a classification label indicating its authenticity. While the raw dataset includes administrative fields such as `id` and `dataset_id`, these were excluded from the analysis as they do not contribute predictive value.

Initial Exploratory Data Analysis (EDA) revealed that the dataset is fairly balanced. It contains 79,995 real news articles (53.7%) and 69,050 fake news articles (46.3%), showing no extreme dominance of one class over the other. Because both classes are well represented, the classification model can learn meaningful patterns from both categories without a heavy bias toward a majority class.

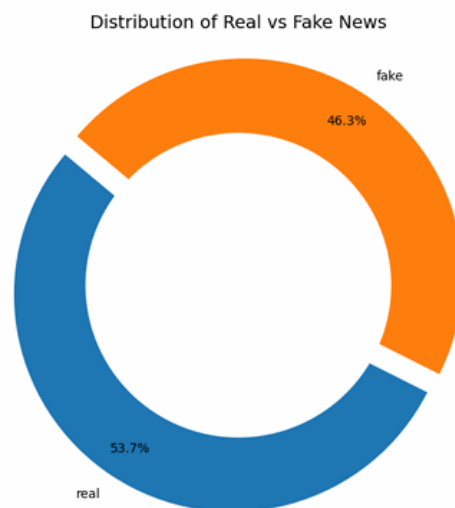


Figure 1: Distribution of Real vs. Fake News within the dataset

1.2 Content Completeness and Article Length

An analysis of content completeness demonstrated that almost all articles in the dataset contain body text. Among the 69,050 fake news articles, 558 have empty content (0.8%). In contrast, out of 79,995 real news articles, only a single article was missing its body text. This strongly suggests that articles lacking body content are highly likely to be fabricated, as professional journalistic standards typically require a body.

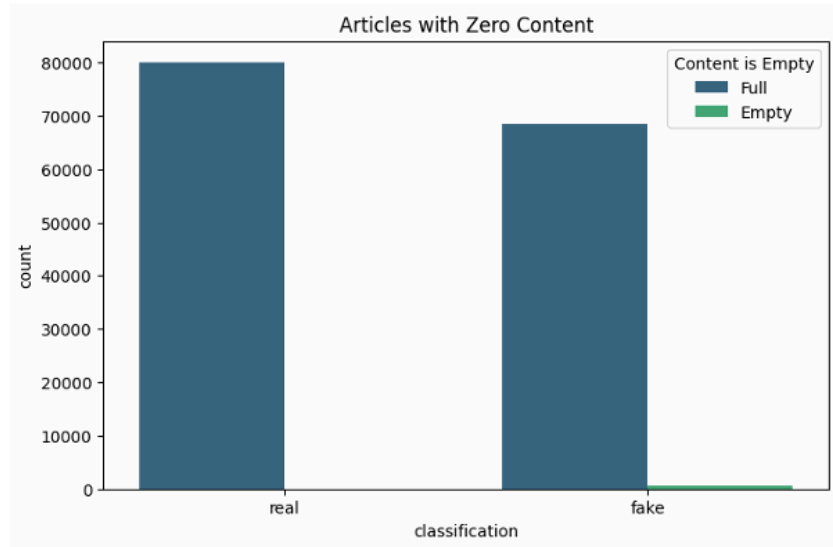


Figure 2: Distribution of articles with zero body content across classes.

Furthermore, a clear disparity in article length was observed. On average, fake news articles are longer, containing approximately 420 words, while real news articles average around 353 words. Despite the presence of noise such as links and repeated characters in the raw text, this average difference of roughly 67 words indicates that total word count is a viable feature for classification.

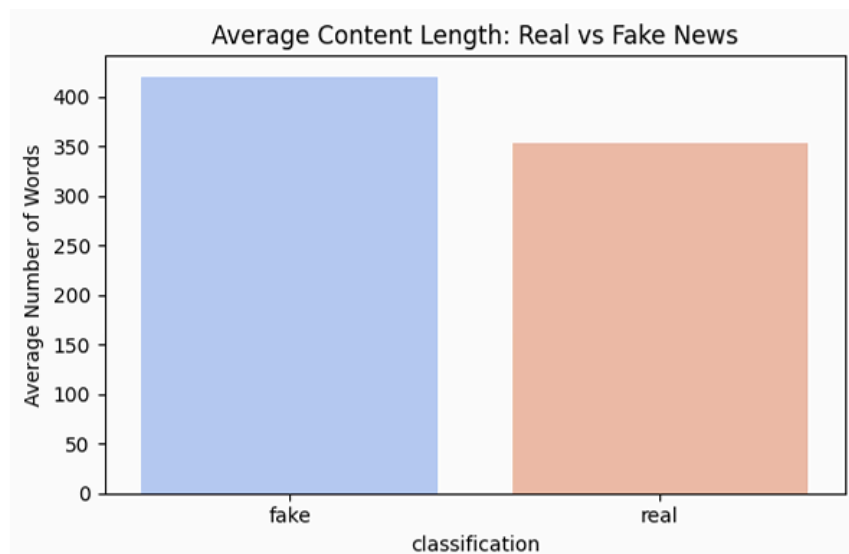


Figure 3: Average content length (word count) of Real vs. Fake News.

1.3 Title-to-Content Ratio

Comparing the ratio of title length to article content length yielded unexpected results regarding common "clickbait" assumptions. Legitimate news articles presented a higher ratio (5.17), indicating that their headlines are longer relative to their concise article bodies. Conversely, fake news articles exhibited a lower ratio (4.77), indicating that their content bodies are disproportionately longer than their titles. This contradicts typical expectations, suggesting that fake news in this dataset relies more on lengthy content bodies rather than purely on attention-grabbing headlines.

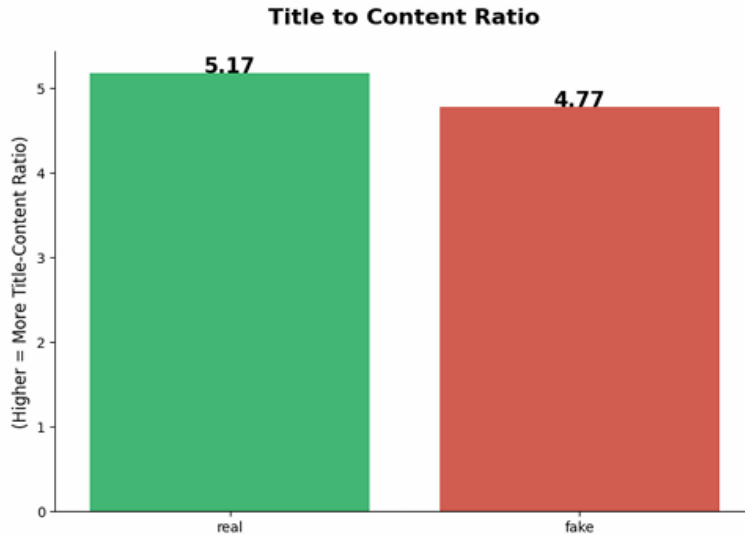


Figure 4: Title-to-Content length ratio comparison.

II. Data Preparation, Cleaning, and Vectorization

2.1 Data Sampling and Text Integration

The primary objective of the preparation phase was to transform the raw text of 150,000 news articles into clean numerical vectors suitable for machine learning algorithms. Before processing the full dataset, a hardware constraint had to be addressed: converting 150,000 rows into a dense matrix for Principal Component Analysis (PCA) required over 3GB of RAM, which caused environment crashes. Consequently, a stratified sample of 5,000 rows (evenly split between Real and Fake) was utilized for visual analysis to maintain statistical significance.

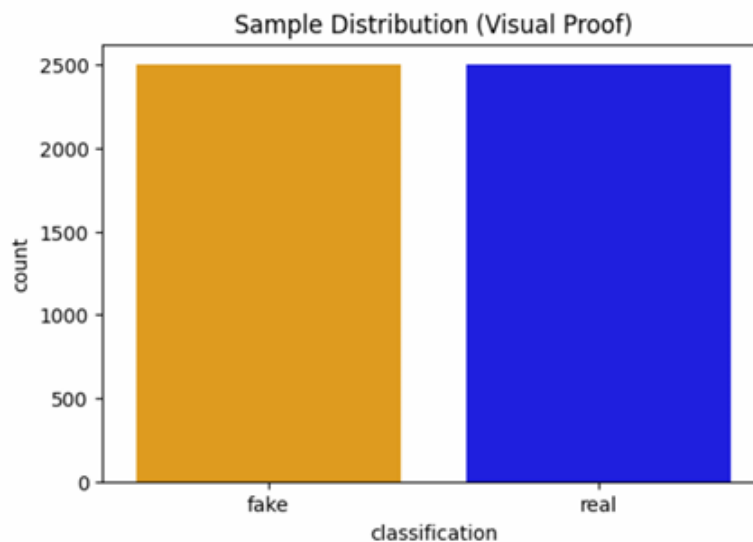


Figure 5: Stratified sample distribution used for exploratory PCA visual proofs.

To ensure crucial signals were captured, the Title and Content columns were merged into a single text field prior to processing. This decision was driven by the observation that fake news often utilizes

sensationalist vocabulary in headlines. Merging the columns ensured these headline signals were unified with the article body, transforming messy titles into learnable features.

2.2 Cleaning Strategy

Two distinct cleaning approaches were evaluated. A baseline "Minimal Cleaning" approach, which only lowercased the text and removed whitespace, was deemed a failure as the data retained significant noise (e.g., HTML tags, erratic formatting) that confused the models.

Subsequently, an "Advanced Cleaning" pipeline was developed. This function lowercased the text, removed punctuation, and replaced entities like URLs and numbers with generic tags (e.g., [Link], [Number]) so the models would learn structural patterns rather than specific websites. The Natural Language Toolkit (NLTK) was utilized to remove stopwords and lemmatize words to group similar conceptual roots. Rows with resulting null values were removed entirely to prevent vectorization errors.

2.3 Vectorization and PCA Analysis

To translate the text into numerical format and visually evaluate the feature engineering strategies, four distinct PCA scatter plots were analyzed.

First, a baseline PCA visualization using Minimal Cleaning was generated. This plot displayed a messy, overlapping blob with no clear groups. This occurred because highly frequent, generic words (e.g., "the", "to") were present in every article, drowning out unique signals and making the classes indistinguishable to the model.

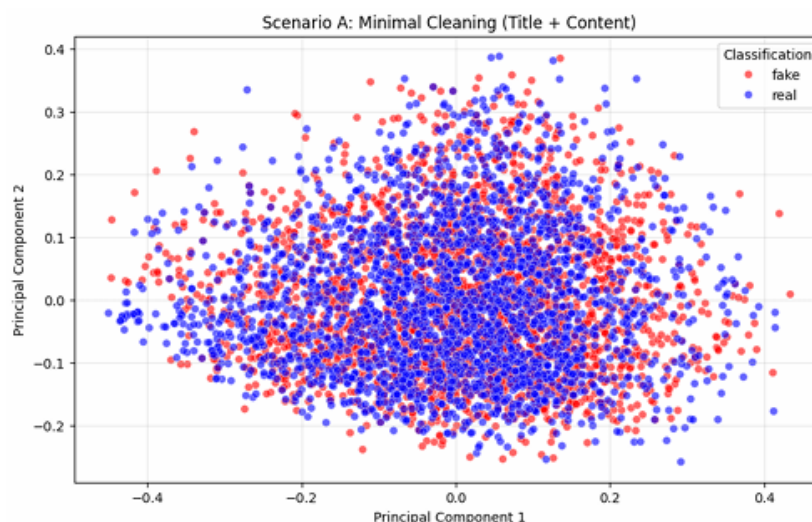


Figure 6: PCA Scatter Plot - Scenario A: Minimal Cleaning.

Second, the Advanced Cleaning pipeline was applied alongside Term Frequency-Inverse Document Frequency (TF-IDF) vectorization. This combination yielded a stark contrast; the PCA graph evolved into a distinct 'V' pattern. This widening visually confirmed that TF-IDF successfully isolated unique "clickbait" vocabulary, pulling fabricated articles away from legitimate news.

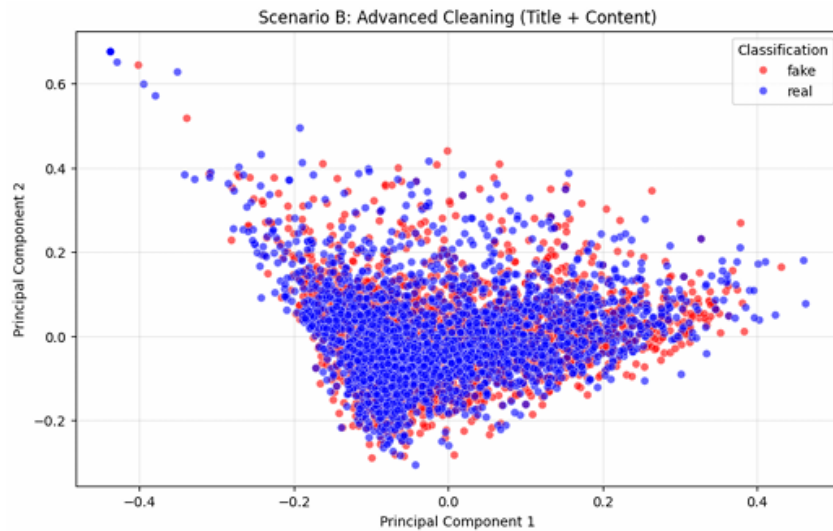


Figure 7: PCA Scatter Plot - Scenario B: Advanced Cleaning (TF-IDF).

Third, the Bag of Words (BoW) technique was evaluated. The resulting PCA plot displayed a tight, dense cluster that confirmed BoW is heavily biased by document length. Short articles were crushed together regardless of their classification, indicating the model grouped data based on text volume rather than semantic meaning.

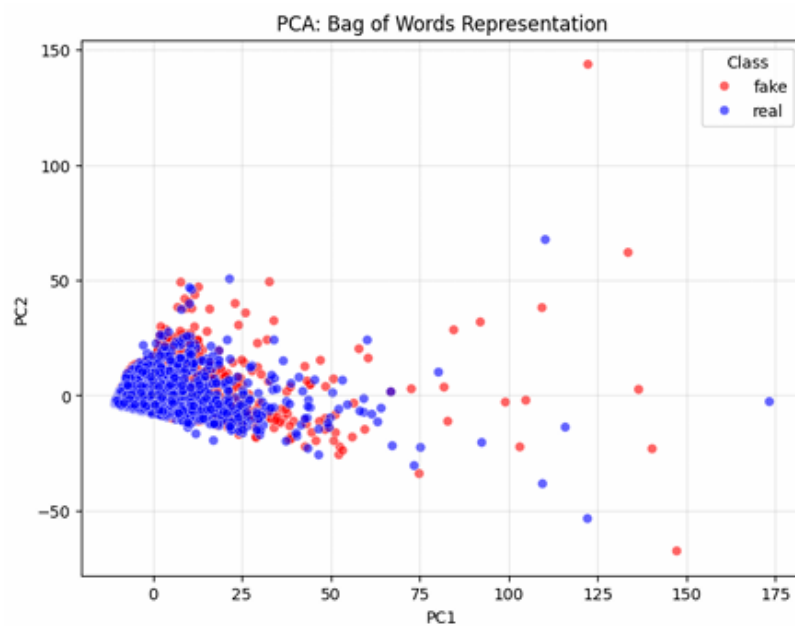


Figure 8: PCA Scatter Plot - Bag of Words Representation.

Finally, Word2Vec was evaluated. The Word2Vec graph showed a dense cluster where Real and Fake data points were heavily mixed. Because Word2Vec averages word meanings into a single vector, the distinct clickbait keywords in the headlines were diluted by the standard vocabulary in the body text.

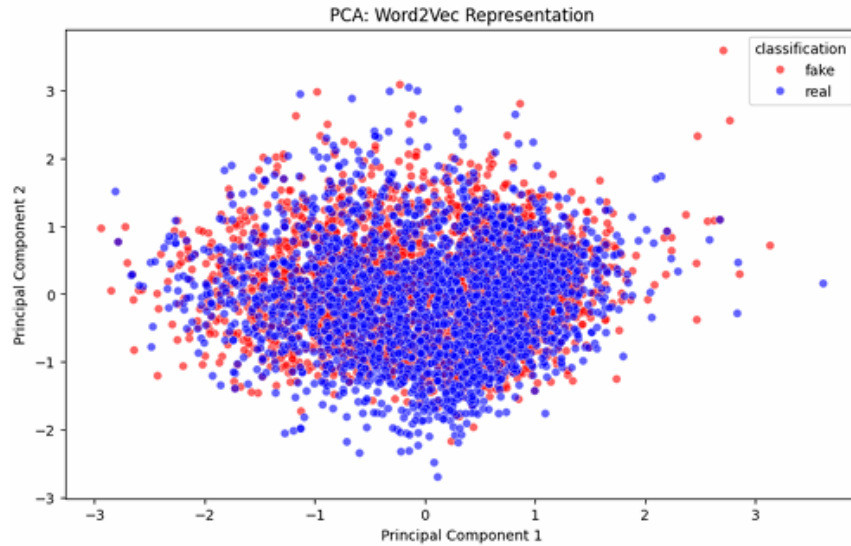


Figure 9: PCA Scatter Plot - Word2Vec Representation.

Variance metrics presented a counter-intuitive finding: Word2Vec captured a massive 45.03% of the variance, while TF-IDF captured only 3.02%. However, because TF-IDF isolated the specific 3% containing the exact features needed to separate the classes visually, both TF-IDF and Word2Vec were carried forward to the final model training phase to test their empirical accuracy.

III. Model Training and Evaluation

3.1 Classification Framework Following the application of the optimized cleaning pipeline to the full training dataset (training_cleaned_full.csv), the study benchmarked three diverse machine learning architectures (Pedregosa et al., 2011).

1. **Logistic Regression:** A standard, efficient model used for binary classification.
2. **Linear SVM:** An algorithm designed to calculate the most effective mathematical boundary separating "Real" and "Fake" categories.
3. **Random Forest:** A more complex ensemble model leveraging multiple decision-making paths.

These models were tested using two distinct feature sets:

- **TF-IDF:** Restricted to the top 3,000 words to focus strictly on the most meaningful terms.
- **Word2Vec (Word Embeddings):** 100-dimensional vectors that mapped every word in the dataset to capture full vocabulary relationships.

3.2 Results and Discussion Model performance was evaluated based on Accuracy Score on unseen test data (test.csv). The results revealed that the vectorization methodology had a more significant impact on accuracy than the choice of the machine learning model itself.

Rank	Vectorization	Machine Learning Model	Accuracy Score
1	Word2Vec	Logistic Regression	55.53%
2	Word2Vec	SVM (Linear)	55.51%
3	TF-IDF	Logistic Regression	50.90%
4	TF-IDF	SVM (Linear)	50.80%
5	Word2Vec	Random Forest	49.22%
6	TF-IDF	Random Forest	48.64%

Table 1: Metric Results of models and vectorization combinations.

Despite TF-IDF providing better visual separation during the PCA phase, Word2Vec consistently achieved higher empirical accuracy across all models during formal training. Word2Vec improved the accuracy of Logistic Regression by 4.63% and Linear SVM by 4.71% compared to their TF-IDF counterparts.

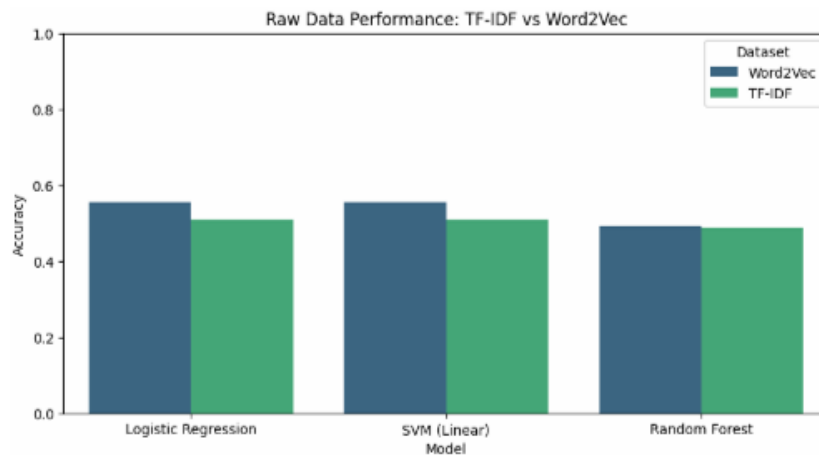


Figure 10: Raw Data Performance comparison between TF-IDF and Word2Vec

Logistic Regression and Linear SVM achieved nearly identical results, representing the strongest predictive outcomes in the study. Conversely, the Random Forest model recorded the lowest accuracy across the entire study, identifying it as the least effective algorithm for this specific data structure.

IV. Conclusion and Limitations

4.1 Conclusion

This study demonstrates that effective preprocessing and vectorization are critical components in the classification of fake news. By merging article titles with body content and applying an advanced text-cleaning pipeline, unstructured text was successfully translated into learnable features.

While TF-IDF effectively isolated clickbait vocabulary for visual separability, Word2Vec's ability to map semantic meaning and word relationships ultimately produced the highest predictive accuracy. Furthermore, simpler linear models outperformed complex ensemble models; Logistic Regression and Linear SVM achieved the best results, peaking at 55.53% accuracy. The most successful configuration for this dataset was the combination of Word2Vec feature engineering with a Logistic Regression classifier.

4.2 Limitations and Future Work Despite identifying an optimal pipeline, the study faces several limitations:

- **Low Overall Accuracy:** The peak accuracy score of 55.53% indicates that the models struggled to confidently distinguish real from fake news, performing only slightly better than a random baseline guess.
- **Feature Constraints:** To manage computational speed and hardware limits, TF-IDF was restricted to 3,000 words and Word2Vec was limited to 100 dimensions, which likely resulted in the loss of nuanced linguistic details.
- **Model Complexity:** The reliance on standard machine learning models, rather than advanced deep learning architectures (e.g., Neural Networks or Transformers), restricted the system's ability to comprehend highly complex writing patterns and context (Chollet, 2026).

Future iterations of this project should address these constraints by incorporating deep learning methodologies and utilizing greater computational resources to process higher-dimensional feature sets.

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